



Society of Petroleum Engineers

SPE-230021-MS

Unlocking Ultra-Sour Gas Reservoirs by Mitigating Elemental Sulphur Deposition While Maintaining Sale Gas Quality

Z. Ahmad, M. Mubarik Albadi, A. Al Hajeri, Mubashir Ahmad, Nnamdi Dike, N. AL SAADI, and S. Ullah Ahmed, ADNOC Onshore; Timothy I. Morrow, ADNOC Research and Innovation Center; A. Ali Khodja, ADNOC Upstream; R. A. Marriott and F. Bernard, Alberta Sulfur Research Laboratory; V. Senich, Schlumberger

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230021-MS](https://doi.org/10.2118/230021-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Unlocking ultra-sour gas reservoirs in brownfields presents many surface and sub-surface operational challenges that must be overcome. Ultra-sour well testing carries risks due to reservoir uncertainties and sour gas handling at surface. During high H₂S reservoir exploration, a novel approach was used to analyze elemental sulfur content in produced solvent samples and validate laboratory results. Various potential elemental sulfur solvent chemicals were tested in the laboratory to determine the most cost-effective choice to prevent elemental sulfur deposition during well testing. A new chemical was evaluated specifically in ultra-sour reservoirs. A custom-built, portable High Pressure Liquid Chromatography (HPLC) apparatus was used for continuous sulfur content analysis in gas and liquid with net zero emissions, and portable H₂S sweetening technologies were assessed.

A specialized oil-based chemical was obtained from the in-country refinery to mitigate surface and subsurface elemental sulfur deposition during ultra-sour gas production. Laboratory tests on samples of the selected chemical indicated improved elemental sulfur solubility compared to diesel over the range of temperature expected to be encountered in the production process. A completion design was chosen to enable downhole delivery of this solvent. Samples of the produced solvent were analyzed at the surface using custom, portable HPLC equipment to determine the optimal solvent injection rate. An advanced ultra-sour gas analyzer was installed at the manifold station to monitor the blended gas and prevent any integrity and corrosion issues while testing the ultra-sour to comply with the blending study. A new portable H₂S sweetening unit technology was also evaluated under various flow conditions.

Introduction

When native sour gas contains very little aliphatic hydrocarbons beyond C₂+, there is the potential to have a natural steady state of elemental sulfur in the reservoir. Upon production, if the native level of sulfur is large, this sulfur can precipitate from the fluid causing restriction in the wellbore and beyond. Evaluating the level of elemental sulfur has evolved over the past 60 years, where bottom hole sampling has been the preferred method since the 1980s. After drilling, test or production wells are cleaned up and samples are

collected with minimal drawdown. Samples are brought to the surface and sulfur content is analyzed after vessel depletion. Sampling vessels must be extremely clean: prior mercury exposure is an issue, thorough solvent rinsing (including CS₂) is required to check for contamination, and oxidants like iron mixing balls must be avoided. Even with strict cleanliness, sample variance can occur if elemental sulfur has already deposited in the wellbore, as tools may pick it up during descent. Despite these challenges, coordinated sampling plans and multiple samplers help ensure reliable results.

Production spans up to three zones with significant elemental sulfur. On-site analysis of returned injected solvent confirmed earlier tests, improved wellbore cleanliness, allowed larger gas samples, and aided mitigation planning. Testing three sulfur solvents verified steady well flow and identified cost-effective options for future use.

The HPHT ultra-sour well with 34 – 36% H₂S was assessed by drilling a vertical pilot hole. These reservoirs intersected by the pilot well, suffered from flow assurance issues due to the presence of elemental sulfur at specific saturation pressure and temperature conditions. During production, solid sulfur deposited within the production tubing at depths between 6000 – 6500 feet, causing flow restrictions as per previous flow testing campaigns.

A range of sulfur solvent chemicals were assessed for their ability to inhibit elemental sulfur deposition, with one chemical demonstrating superior solubility and cost-effectiveness. Downhole injection is scheduled at a rate of 42–80 litres per minute, accompanied by continuous sampling of the produced sulfur solvent to monitor dissolved sulfur concentrations and facilitate real-time optimization of the solvent injection rate. This approach is necessary because: (i) although certain sulfur solvents may be readily available for production, robust supply for flow testing is inconsistent, necessitating the use of diesel in larger quantities; (ii) solvent return can indicate active sulfur deposition, requiring adjustments to solvent-to-gas flow rates; and (iii) when the solvent is undersaturated with sulfur, consistent sulfur loading in the gas should be observed across varying solvent-to-gas flow rates. Due to logistical constraints, diesel was utilized as the sulfur solvent in one zone, while two additional zones underwent comparative analysis using a specialized sulfur solvent chemical during field trials. The new chemical exhibited fivefold greater effectiveness in sulfur inhibition compared to diesel.

At Well-X an ultra-sour gas analyzer was utilized during well testing for continuous gas monitoring, blending sour gas with sweet gas to mitigate any downstream flow assurance issues. This online analyzer supported the testing of the ultra-sour gas well at various rates by close monitoring of the blended gas. A new portable sweetening technology was evaluated under different flowing conditions to manage the ultra-sour gas, with detailed techno-commercial studies conducted for future application. Flow limitations were concluded to be due to the elemental sulfur handling capacity of 20 metric tons per day.

Completion and surface equipment design

The well-X was drilled in 2025 with L-80 4-1/2" completion. During the completion design phase, multiple injection rates were evaluated for downhole injection, ranging from 1.5 liters per minute to 45 liters per minute. Capillary injection was deemed unsuitable due to the high injection rate required for managing elemental Sulphur. The completion was equipped with an SPM to facilitate the use of either a chemical injection valve or a circulation valve, incorporating specialized sealing elements designed to accommodate elevated injection rates. This strategy presented limitations regarding precise monitoring of injection rates, which was verified by tracking fluid returns at the surface.

Specialized injection pumps were procured to meet both low and high injection requirements for diesel and the new chemical "L Solvent". At the surface, to mitigate elemental Sulphur under worst-case scenarios, a debris catcher with a specific mesh size was implemented to prevent sulfur accumulation in the separator and flow line.

Sampling Procedure

Collecting surface diesel samples in dry gas environments presents challenges for obtaining representative samples necessary for field trial conclusions. Separator samples were gathered from the oil level glass side to improve diesel sample quality ([figure-1](#)). Following each dosage, it was necessary to drain the separator according to the previous injection rate to ensure complete removal of diesel. Before changing the choke or adjusting the diesel injection rate, all accumulated liquid in the separator was drained. The well was allowed to flow for 2–3 hours to collect an adequate amount of injected diesel sample (3–5 ml) for analysis. Additionally, some samples were collected and analyzed directly from the choke manifold for comparison.



Figure 1—diesel samples at site

Mitigation of Elemental Sulfur Deposition

[Figure 2](#) shows typical sulfur solvency based on a previous review of available literature. While water has very little sulfur solvency, polycyclic aromatic solvents outperform most physical solvents. The diallyl disulfide (DADS) and dimethyl disulfides (DMDS) on the righthand side of [figure 2](#) are chemical solvents, which incorporate sulfur in the molecules itself. It is often difficult to secure consistent supply for these solvents and/or they can be extremely difficult to handle. For example, DMDS can release thiols. The disulfides are typically used for batch treatments but can be used in continuous injections. Carbon Disulfide (CS_2) has high vapor pressure and is rarely used. As such, physical sulfur solvents (and amines) are mostly used in commercial production. They can be deployed by batch soaking systems (bull heading wellbores) or by continuous injection. Note that for either mitigation strategy, it is the lowest temperature in the system that is important, as the objective is to keep sulfur from depositing anywhere in the production system.

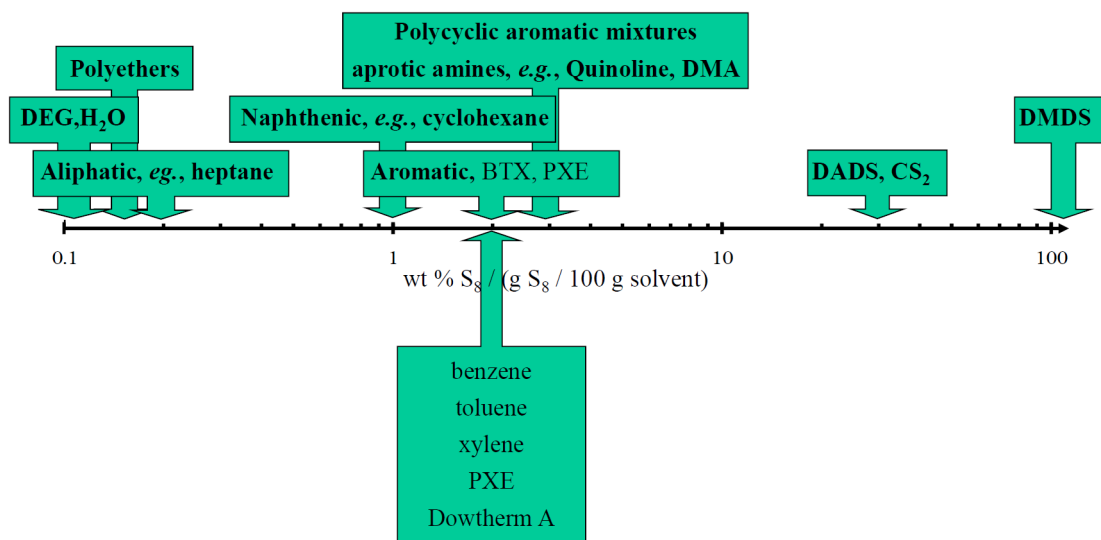


Figure 2—Typical solvency for generic sulfur solvents.³ Note that DADS and DMDS are chemical solvents and amine hydrosulfides are not shown

Typically sulfur solvents are analyzed by complexing triphenylphosphine and then following the sulfide by GC (gas Chromatography)/ Pulsed flame photometric detector (PFPD). If no chemical uptake is expected, HPLC can be used, as long as there is sufficient separation in the chromatography.

In 2022, a laboratory screening study of six different elemental sulfur solvents was conducted. Diesel, aromatic naphtha, and a specialized oil-based product from the refinery were among the solvents testing in this study. The study identified aromatic naphtha and the specialized refinery oil product as top-of-class solvents, with both solvents having a sulfur capacity exceeding 40 grams of sulfur per liter of solvent at a temperature of 60°C. Diesel was determined to be a comparatively poor sulfur solvent, with the sulfur capacity at 60°C measuring about 4-5X lower than the top-of-class solvents.

The specialized refinery oil product, referred to herein as "Solvent L", was determined to be the more cost-effective than aromatic naphtha. New samples of Solvent L from the refinery as well as a 3rd party-supplied alternative "Solvent L*" were provided in 2024 for elemental sulfur capacity testing. Photographs of samples of these three sulfur solvent candidates are shown in Figure 3. Note that while Solvent L has superior uptake, it also has low viscosity, which is beneficial. A high viscosity can limit the use of the solvent.

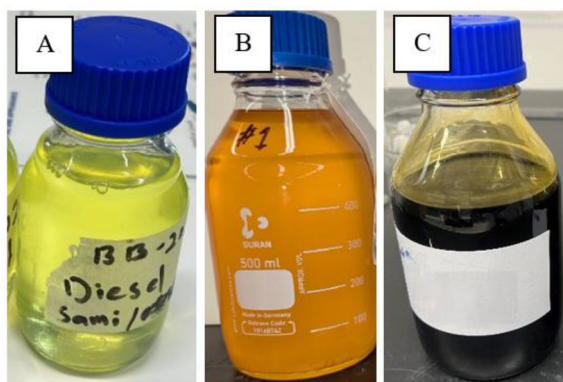


Figure 3—Candidate Sulfur Solvents: (a) diesel, (b) Solvent L, (c) Solvent L*

ASRL (Alberta Sulphur Research Ltd) performed compatibility testing with this solvent with (i) used drilling mud, (ii) unused drilling mud and (iii) corrosion inhibitor. The results showed no issues. It should

be noted that sulfur solvents themselves, if undersaturated, will mitigate corrosion, because under-deposit sulfur contact corrosion is aggressive.

The basic procedure used for measuring the elemental sulfur capacity of each solvent versus temperature was to dissolve annealed sulfur powder into the solvents on a stirring hotplate at 25, 40 and 60°C. Periodically, the stirring was halted, and aliquots were removed for analysis. If there was no visible sulfur in the vessel, more was added. These aliquots were analyzed by HPLC.

The laboratory-measured elemental sulfur capacities, in grams of sulfur per liter of solvent, for refinery Solvent L, 3rd party-supplied Solvent L* at 25°C, 40°C, and 60°C are shown below in Figure 4. Figure 4 also shows the estimated sulfur capacity versus temperature of diesel as calculated from an empirical curve fit to experiments that were performed in the 2022 screen study.

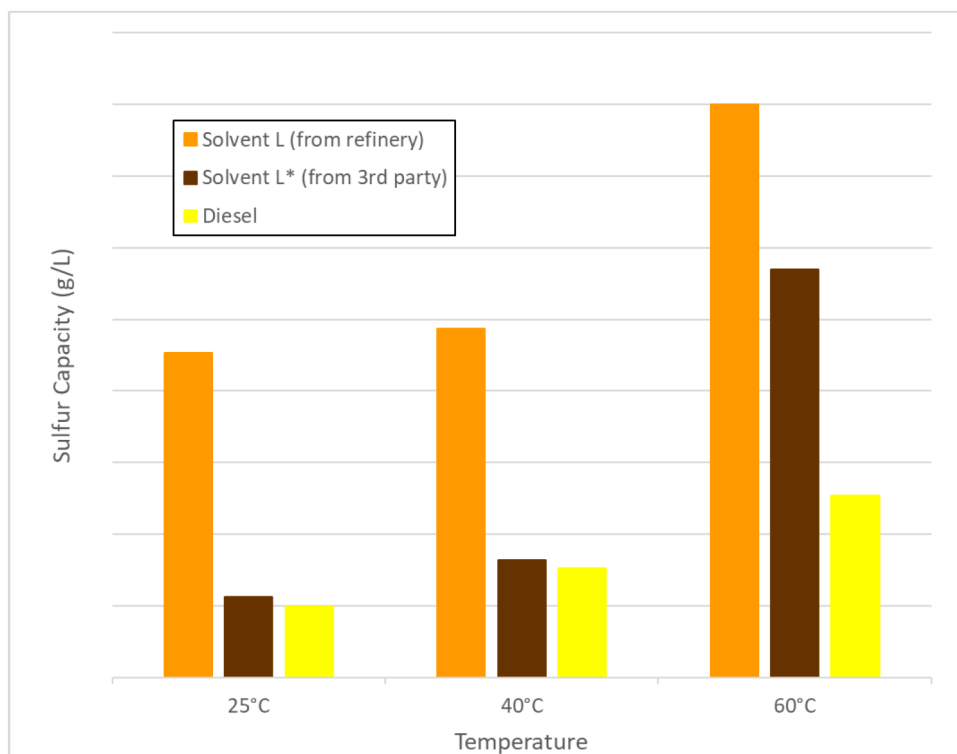


Figure 4—Dissolved elemental sulfur capacity of solvent options versus temperature.

It can be observed from Figure 4 that Solvent L outperforms the 3rd-party supplied Solvent L* by ~40% at 60°C and can hold 200 – 300% more sulfur than Solvent L* at 40°C and 25°C, respectively. Solvent L also shows about 3 - 5 times greater sulfur uptake than is expected from diesel. The alternative Solvent L* also proved difficult to extract through syringe filters, suggesting that this solvent contained some unmeasured quantity of suspended solid particles or is too viscous. Based on these results and cost-effectiveness considerations for each solvent, the decision was made to use refinery Solvent L as the preferred method for inhibiting sulfur during well testing, with diesel to be used as a back-up solvent in case of any issues with getting the required volumes of Solvent L to the well site.

For on-site analysis of continuous sulfur solvent injections, ASRL built a bespoke and simplified HPLC system which could be transported to site and rapidly deployed from the labs. The instrument can measure the sulfur in the returned solvent, where sulfur dewpoint (precipitation temperature) can be compared to gas flow rate and sampling conditions. This comparison was done remotely from Calgary, whereas field staff operated the instrument in the field. As some iron sulfides and other contaminants can be captured, all samples were centrifuged and sent through a 0.25 µm filter. Solvents can be captured at the three-phase

test separator or ideally at the choke manifold. Note that this analysis requires accurate knowledge of gas flow and solvent flow.

A photo of the custom portable HPLC apparatus is shown below in Figure 5.

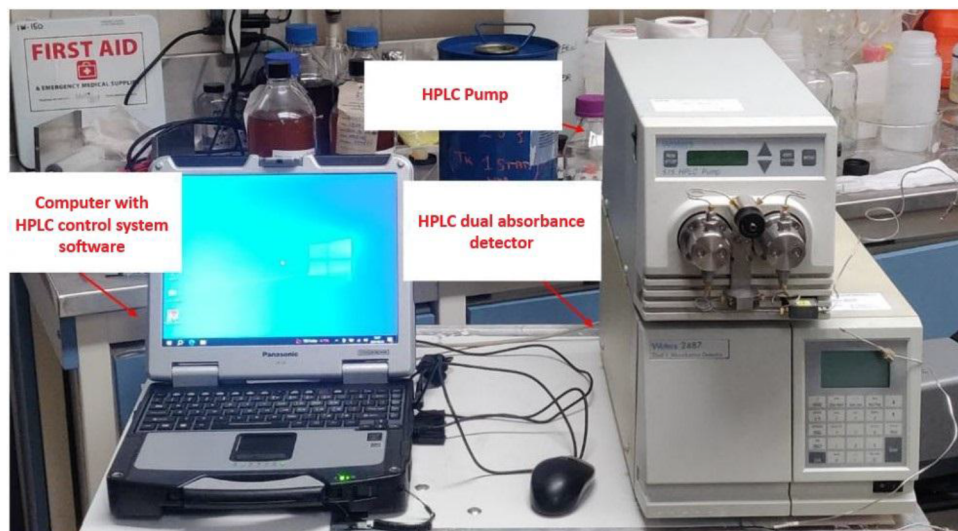


Figure 5—Custom portable HPLC (High Pressure Liquid Chromatography) setup for on-site sample analysis.

Solvent Performance During Flow Testing

Well-X intersected three distinct ultra-sour gas intervals. Each interval was tested individually after isolation, perforation, and acid-washing operations. The first interval tested was the lowermost flow unit, referred to herein as Interval 1. Based on reservoir data from offset wells, Interval 1 was not expected to suffer from significant quantities of elemental sulfur deposition (although past tests did show enough sulfur for deposition). The flow test for Interval 1 was planned to use a gas production rate of 8-10 MMSCFD. For this gas rate, the optimum pumping rate of Solvent L was predicted to be well below the minimum injection rate of the on-site chemical pump. Diesel was made available onsite instead of Solvent L for the production test of Interval 1.

The flow test for Interval 1 was planned to use a gas production rate of 10 MMSCFD. For this gas rate, the optimum pumping rate of Solvent L was predicted to be well below the minimum injection rate of the on-site chemical pump. Diesel was made available on site instead of Solvent L for the production test of Interval 1.

The gas production rate during the Interval 1 production test was stable around 10 MMSCFD. Initially, the diesel injection rate into the downhole chemical delivery string was 5 bpd. Four separate produced diesel samples were collected from a sampling point at the gas-liquid separator over a period from 9AM to 1:30PM (roughly 60 – 90 minutes apart). On-site HPLC analysis of these four diesel samples showed consistent dissolved sulfur levels that were below the saturation capacity of the diesel. After analysis of the fourth sample, the decision was made to reduce the diesel injection by 50% to 2.5 bpd. Three additional produced diesel samples, all collected after the reduction in diesel injection rate, all showed consistent quantities of dissolved sulfur that were safely below the saturation limit for diesel. The production test for Interval 1 was successfully concluded without any deposition of elemental sulfur.

The wellhead temperature during these tests was greater than 75°C (168 F) and the test separator was at approximately 50°C (120 F). Initial returned diesel samples showed a saturation temperature at -30°C (0.21 bbl h⁻¹ solvent and 10 MMScf d⁻¹ gas). Upon adjusting the injection rate, the saturation temperature was raised to -10°C, which still allowed for full sulfur transfer to the solvent, i.e., the saturation temperature was less than the test separator temperature where most samples were captured. With a final solvent flow of 0.10 bbl h⁻¹ and gas production rate of 6.9 MMScf, the diesel contained 1.6 ± 0.1 g L⁻¹ elemental sulfur. The latter

indicates that the gas had 0.20 ± 0.01 lbs MMScf⁻¹ of elemental sulfur. This measurement was reproduced after draining the test separator and also through a choke manifold sample (instantaneous sample). Note that less than 5 mL of sample is required for this analysis.

A sulfur saturation phase diagram for this fluid is shown in Figure 6, where the 0.2 lbs MMScf⁻¹ is reported. The phase diagram indicates that some deposition would have occurred after the choke and before the separator, if the solvent was not deployed. However, deposition would not have occurred in the wellbore. It should be noted that this is a very low number for bottomhole analyses, considering potential oxidation and the limited gas sample volumes. Previous bottomhole samples for this zone showed 8 ± 2 lbs MMScf⁻¹ sulfur (2014, 2018 and 2020). This highlights the utility of the solvent return analysis. Much lower numbers are measurable, as long as the solvent is undersaturated at the sampling location. The phase behavior is an updated ASRL model.

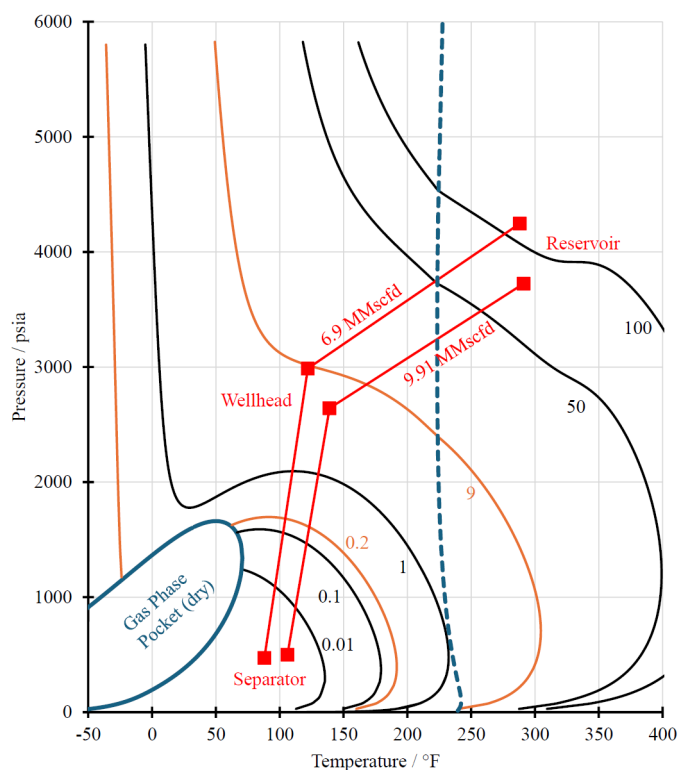


Figure 6—A sulfur – sour gas phase diagram showing several sulfur phase pockets for the initial interval flow test. Solvent return showed 0.2 lb MMScf-1 sulfur, whereas bottomhole samples showed 9 lb MMScf-1 sulfur.

Roughly 6 weeks later, a flow test of Interval 2 was performed. Based on historical data from offset wells, Interval 2 was believed to be at substantial risk for elemental sulfur deposition. It was intended to inject Solvent L to mitigate the sulfur deposition risk for this flow test, but for logistical reasons Solvent L was not available on-site and diesel again had to be deployed as the sulfur solvent. During the flow-to-clean period after an acid wash, the gas flow rate stabilized at ~5 MMSCFD and on-site analysis was performed of returned diesel samples. The diesel was injected at 14.4 bpd, which was the maximum rate possible from the on-site chemical pump. On-site HPLC testing showed the returned diesel to be fully saturated with elemental sulfur, suggesting that sulfur deposition might be occurring somewhere upstream of the separator, possibly in the production tubing and choke. As the diesel injection rate could not be increased any further, it was decided to choke back the gas production rate to 3.8 MMSCFD. However, diesel samples produced at this higher solvent-to-gas ratio were still fully saturated. As a result, a higher capacity pump was brought to site which increased the diesel injection rate to ~400 bpd.

Tubing clearance and bottom hole sampling conducted post first trial of diesel injection. Tubing was no clear from any deposition and decided to test well at various gas rate with higher diesel injection rates. As per lab testing, elemental Sulphur showed solubility in diesel as per below table;

Table 1—diesel injections at various gas rates

Assumption ($T_{min} = 60\text{ }^{\circ}\text{C}$):	Worst Case Scenario	Best Case Scenario
S ₈ Content* (lbs MMscf ⁻¹)	200	8
Gas Production Rate (MMscf/d)	Injection Rate for Diesel (L/min)	Injection Rate for Diesel (L/min)
5.00	25	1.0
10.00	50	2.0

While sampling during operation injection was carried out at 45, 90 lit/min but still sample was observed saturated with elemental Sulfur.

On July 31st, a larger capacity pump was installed; however, there were technical issues with the diesel injection, which caused injection rates to be very low. The solvent saturation temperature was within 1°C of the reported separator temperature, indicating active deposition during this flow test. We note that solid sulfur samples were being captured at the choke, indicating full agreement with the analytical results. In Late August, solvent L will be deployed at the lower injection rate. This will be a significant milestone for producing the fluid from this zone, which had previously been too expensive to develop.

In August, the same interval was re-tested with diesel again to reverify the results after making sure functionality of chemical injection valve and injection pump up to mark. Well successfully tested well at 5 and 10 MMSCF/D flow with 16.8 bbl/hr injection and observed all diesel samples undersaturated with elemental sulfur. In Late August, solvent L will be deployed at the lower injection rate (3 – 4 times lesser). This will be a significant milestone for producing the fluid from this zone, which had previously been too expensive to develop.

Ultra Sour gas flow by maintaining sales gas quality

Ultra sour well testing conducted containing 32-mole% of H₂S with 10 mole% of CO₂ by utilizing available less Sour gas (3-4%). The Objective of this activity is to explore maximum potential of the proposed wells in conjunction with observing sour gas blending effect with low H₂S well fluids from existing reservoirs.

Initial blending estimates based on Compositions and gathering stations production profiles suggest that Up to ultra Sour 10-12 MMSCFD likely to be mixed at gather station while maintaining H₂S limit of <10 mole% at existing production manifolds / respective trunk lines of depletion reservoirs and <7 mole% at customer battery limit .

Proposed sour gas blending ratios are calculated in such a way so that net effect of H₂S at compressor station common suction header (post mixing) will remain < 10 mole%, any increment shall be managed through negotiating production from less sour wells.

A suitable wellhead testing package, including all necessary safeguarding measures and utilities such as hydrate inhibition, is to be provided by the company. The gas will flow directly to the gathering station depletion trunk lines, with appropriate controls and valves installed at the wellhead as part of the well testing package. The study scope includes the tie-in of candidate well flow lines from downstream of the well testing package to the production manifold slot. Corrosion mitigation has been considered for the use of existing materials and equipment in environments with H₂S concentrations of approximately 32 mole% during short-duration well tests. To maintain gas quality, elemental sulfur and other sulfur species are not permitted to be mixed with system gas.

Table 2—blending study results

Ultra-Sour flow (MMSCF/D)		Total flow	H ₂ S %	CO ₂ %
12	Gathering Station	95.82	9.24	6.76
	Compressor station			
12		267.5	7.29	6.69

Ultra Sour Gas Analyzer

Sour gas well testing can be conducted without flaring by blending available production gas. At well X, a temporary carbon steel flow line was installed to support three months of testing, connecting the well to a nearby gathering station. A flow of 12 MMSCF/D was blended with 83 MMSCF, resulting in an H₂S concentration of 9.24%. This gas was further blended at the compressor station to achieve sales gas quality.

At the gathering station, a specialized portable H₂S analyzer was deployed to provide real-time measurements of H₂S and gas composition up to C₆+ components, with a sampling interval of five minutes. The analyzer demonstrated 99% accuracy when analyzing untreated separated gas and was used to manage ultra-sour flow throughout the testing period.

Portable H₂S Gas Sweetening Technology for Remote Natural Gas Processing

The growing demand for natural gas and the discovery of new reserves in remote locations have highlighted the need for portable and mobile gas desulfurization units. Traditional gas processing infrastructure is often unavailable in these areas, making it difficult to utilize natural gas without releasing harmful sulfur compounds. A new portable H₂S gas sweetening unit, capable of treating 20 tons of sulfur per day, has been evaluated to address this challenge. The unit removes H₂S from natural gas to levels safe for flaring (without SO₂ formation) or pipeline reinjection.

Key Features:

- Portable & Temporary Installation – Rapid deployment, commissioning, and decommissioning.
- Low Operating Costs – Minimal chemical consumption and energy-efficient operation.
- Safe & Reliable – Ensures consistent treatment while reducing environmental risks.

Process Overview ([Figure 1](#)): Aqueous solution utilises a continually regenerated chelated iron catalyst to convert H₂S to elemental sulfur. The sulfur is filtered from the solution and can be used as fertilizer or disposed of in landfill. Chemicals and process has minimal HSE concerns.

There are two chemical reactions in the process:

- Reaction of H₂S in Absorber (contactor) producing elemental sulfur
- Reaction of atmospheric oxygen in the oxidizer to regenerate the solution

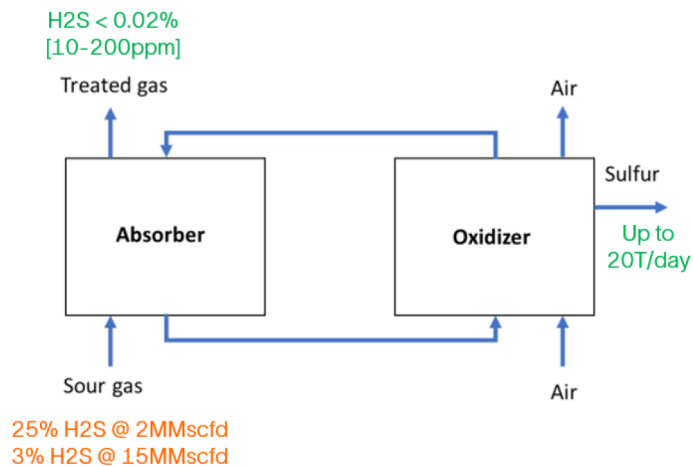


Figure 8—Principal working diagram

The chemical reactions that take place during the process:

1. Reaction of H₂S with Fe³⁺ in Absorber (contactor) with the product of elemental Sulphar

$$2 \text{Fe}^{3+} - \text{EDTA} + \text{H}_2\text{S} = 2 \text{Fe}^{2+} - \text{EDTA} + 2 \text{H}^+ + \text{S} \text{ (reaction in contactor)}$$
2. Reaction of atmospheric oxygen and Fe²⁺ in oxidizer

$$4 \text{Fe}^{2+} - \text{EDTA} + \text{O}_2 + 4 \text{H}^+ = 4 \text{Fe}^{3+} - \text{EDTA} + 2 \text{H}_2\text{O} \text{ (reaction in oxidizer)}$$

Below figures 9&10 show operating and efficiency envelopes of the unit.

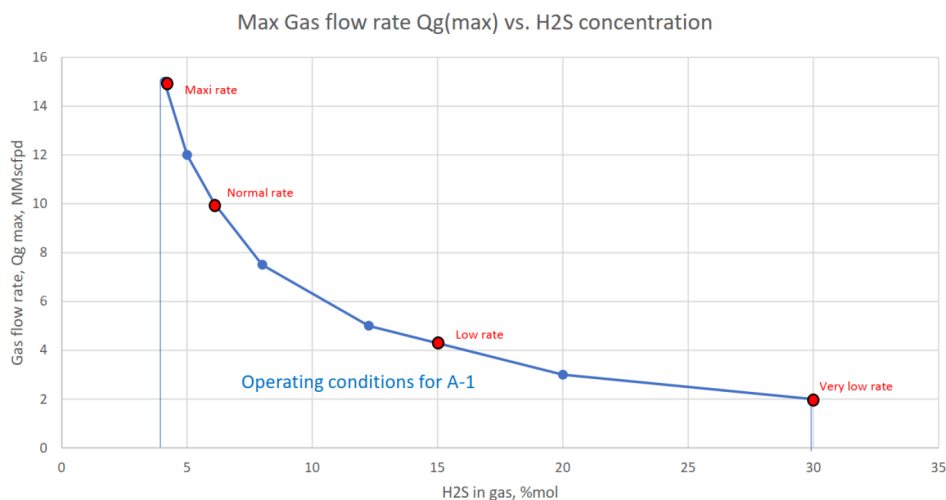


Figure 9—H₂S gas sweetening unit operating envelope

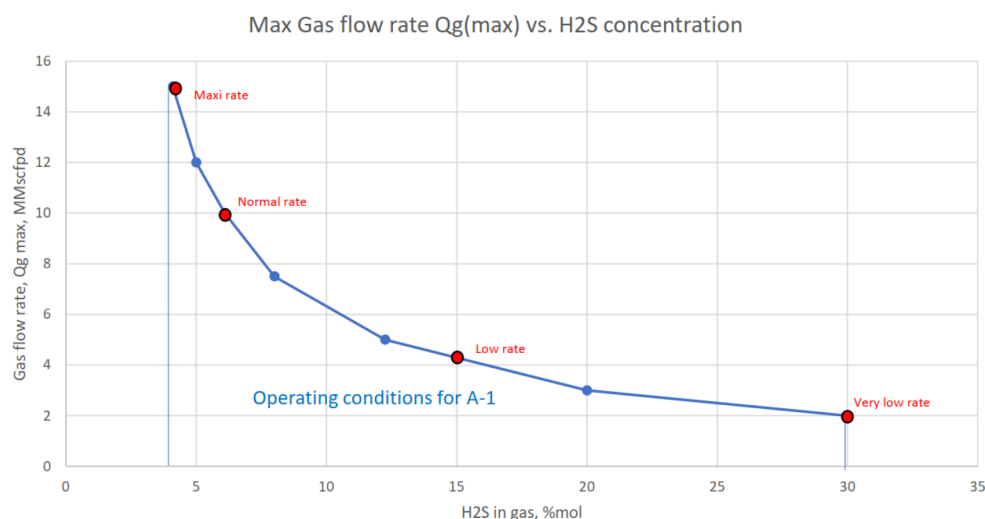


Figure 10—H₂S gas sweetening unit efficiency envelope

Typically, the cost of treating one ton of Sulphur with these portable H₂S sweetening units is 3–4 times lower than with direct H₂S scavengers, thanks to continuous chemical regeneration.

In summary, the evaluation confirms that this portable H₂S gas sweetening unit offers a cost-effective and sustainable solution for natural gas sweetening. It enables safe, efficient processing of sour gas for compliant flaring or pipeline-ready transport.

Conclusion

A specialized oil-based chemical, "Solvent L," from refinery demonstrated superior elemental Sulphur solubility and cost-effectiveness compared to diesel and other solvents, showing 3-5 times greater Sulphur uptake than diesel. This solvent was chosen for downhole injection during well testing to inhibit Sulphur deposition. Laboratory tests measured Sulphur capacity of solvents at various temperatures using HPLC analysis. Field trials on different reservoir intervals showed that diesel was insufficient at higher Sulphur loads, while Solvent L allowed better Sulphur management, optimizing injection rates based on continuous sampling. A portable high-pressure liquid chromatography device was developed for on-site analysis of Sulphur solvent samples, enabling real-time monitoring of dissolved Sulphur content to adjust solvent injection rates effectively during well testing. The well was completed with equipment supporting high-rate chemical injection, including specialized valves and pumps, and surface debris catchers to prevent Sulphur accumulation, ensuring flow assurance during ultra-sour gas production. To maintain sales gas quality, ultra-sour gas containing up to 34% H₂S was blended with less sour gas to keep H₂S concentrations below pipeline limits, using real-time gas analysis and corrosion mitigation strategies during short-duration well tests. A portable ultra-sour gas analyzer provided continuous, accurate measurements of H₂S and gas composition at the gathering station, facilitating management of sour gas blending and flow assurance without flaring. A new mobile gas sweetening unit was evaluated, capable of treating 20 metric tons of Sulphur per day by converting H₂S to elemental Sulphur using a regenerated chelated iron catalyst, offering a cost-effective and environmentally safe solution for remote sour gas processing. Flow limitations during testing were linked to elemental Sulphur handling capacity. Adjustments in solvent injection rates and gas production flows were necessary to prevent Sulphur saturation and deposition, ensuring well integrity and production efficiency.

References

1. R. A. Marriott, P. Pirzadeh, J. J. Marrugo-Hernandez and S. Raval (2016) Hydrogen sulfide formation in oil and gas, Canadian Journal of Chemistry, *Can. J. Chem.* **94**, 406-413.
2. J. B. Hyne (1968) Sulphur Deposition in Sour Gas Wells. *ASRL Quarterly Bulletin*, Vol. **V-1**, 1-21.
3. R. A. Marriott, P. M. Davis and P. D. Clark (2007) Estimation of sulfur deposition during the production of lean sour gas, Proceedings of the 3rd International Conference on Sour Oil & Gas Advanced Technology, Abu Dhabi, April 2007, Dome Exhibitions, UAE, 2007, 121-140.
4. P. D. Clark and K. L. Lesage (1989) Quantitative determination of elemental sulfur in hydrocarbons, soils, and other materials, *J. Chromatographic Sci.* **27**, pp. 259-261 (1989).